

Garance Malnoë - Curriculum Vitae
M.Sc Student in Statistics and Data Science

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EDUCATION

Sept 2024 - July 2026 | M.Sc. (Master) in Statistics and Data Science

Université Grenoble Alpes

- Ranked 1st of my class for Year 1 and Year 2 (Semester 1).

Sept 2021 - May 2024 | B.Sc. (Licence) in Fundamental Mathematics

Université de Rennes (Years 1 & 2) and École Normale Supérieure (ENS) Rennes (Year 3)

- Advanced coursework in applied and fundamental mathematics.
- Ranked 1st of my class in Years 1 & 2.
- Admitted to an excellence program (*Grande École*) in Year 3 via a selective national competitive exam (*concours magistère*).
- Minor in Computer Science.

Sept 2018 - June 2021 | Baccalauréat (high school diploma)

Lycée Joliot-Curie, Rennes

AWARDS & SCHOLARSHIPS

2025 | Excellence Scholarship - Multidisciplinary Institute in Artificial Intelligence (MIAI)

- Merit-based scholarship awarded to highly ranked students studying Artificial Intelligence, for the 2nd year of the M.Sc. program.

2024 | Mitacs Globalink Research Internship Award

- Competitive international grant fully funding a 12-week research internship at a Canadian academic institution.

RESEARCH EXPERIENCE

March - July 2026 | Research Intern : “Sequence analysis for life trajectories: a comparison of R and Python environments and the integration of AI methods”

TIMC and LIG laboratories, Université Grenoble Alpes, France - Supervised by Caroline Bazzoli (TIMC) and Jérôme Gensel (LIG)

Nov 2025 - Feb 2026 | M.Sc. Year 2 Project : “Measuring the quality of idea extraction performed by a large language model: application to public consultations with open-ended questions”

Data For Good - Supervised by François Cyril

- Evaluated LLM performance on text data from French public consultations, comparing metrics from literature (ROUGE, NLI, QualIT) with human grading.
- Developed a Python-based pipeline using Ollama to automate extraction and grading of ideas from open-ended questions.
- Presented our methodology and results to an academic jury alongside two Master's students.
- **Links:** [\[Report \(french\)\]](#) [\[Oral presentation support \(french\)\]](#)

May - Aug 2025 | Mitacs Research Intern in Canada : “How to classify responses to adversity: resilience and vulnerability as deviations from the average”

Resilience Research Centre, Dalhousie University, Halifax, NS, Canada - Supervised by Jan Hölte

- Conducted a literature review and adapted existing methodologies and designed new approaches for resilience research.
- Implemented classification methods in R to categorize adversity responses into ‘resilient’, ‘non-resilient’ and ‘normative’ profiles across cross-sectional and longitudinal real-world data.
- Drafted the methods and results sections as a co-author for a scientific paper currently in preparation.
- Presented intermediary research methodologies and findings to an academic jury.
- **Links:** [\[Draft \(english\)\]](#) [\[Oral presentation support \(french\)\]](#)

May - June 2024 | Research Intern : “Reconstruction of origin-destination matrices based on passenger flows”

LMO, Université Paris-Saclay, France - Supervised by Bertrand Maury and Sylvain Faure

- Applied optimization and statistical techniques to reconstruct origin-destination (OD) matrices from Lagrangian data.
- Implemented these mathematical approaches in Python on real-world data.
- Presented intermediary research methodologies and findings to an academic jury.
- **Links:** [\[Report \(french\)\]](#) [\[Oral presentation support \(french\)\]](#)

Jan - April 2024 | B.Sc. Year 3 Project : “Epidemic dynamics modelled using ordinary compartmental models and ordinary differential equations”

IRMAR, Université de Rennes, France - Supervised by Hugo Martin

- Studied Ordinary Differential Equations (ODEs) theory and compartmental models to modelize epidemic dynamics alongside another 3rd year student.
- Implemented Python simulations to visualize theoretical epidemiological evolutions.
- **Link:** [\[Report \(french\)\]](#)

May - June 2022 | Research Intern

IRMAR, Université de Rennes, France - Supervised by Nathalie Krell

- Studied paradoxes in probability (e.g. Monty Hall problem) : developed theoretical demonstrations, modeled results in Python and designed and conducted a small survey alongside another 1st year student.

PUBLICATION

Malnoë, Hölte et al. *“Resilient, Non-Resilient, or Normative? A Methodological Take on What Counts as a Better-Than-Expected Outcome in Resilience Research.”* Manuscript in preparation (Expected submission: 2026)

INDUSTRY EXPERIENCE

May - August 2023 | R&D Intern

Orange, Rennes, France

- Designed and implemented independently a Python-based Machine Learning tool to predict optical-fibre installation failures.

METHODS & TOOLS

Programming languages: Python and R (Proficient / Daily use) | Julia and SQL (Working knowledge)

Core statistics: Parametric/non-parametric estimation, generalized linear models, bootstrapping methods, and dimension reduction methods.

Advanced statistical modelling: Mixed models, Bayesian statistics, survival analysis, time series, and sequence analysis.

Machine learning and high-dimension data: Supervised/unsupervised learning, high-dimensional modelling, Natural Language Processing, and deep learning.

Data Visualization & Dashboards: Interactive dashboard development with Python (Dash and Plotly), and PowerBI.

PERSONAL ENGAGEMENT

2023 - 2024 | Coordinator, "Young Women in Math & Computer Science" (RJMI)

- Organized a 3-day research camp to encourage high school girls to pursue studies in mathematics and computer science at ENS Rennes.